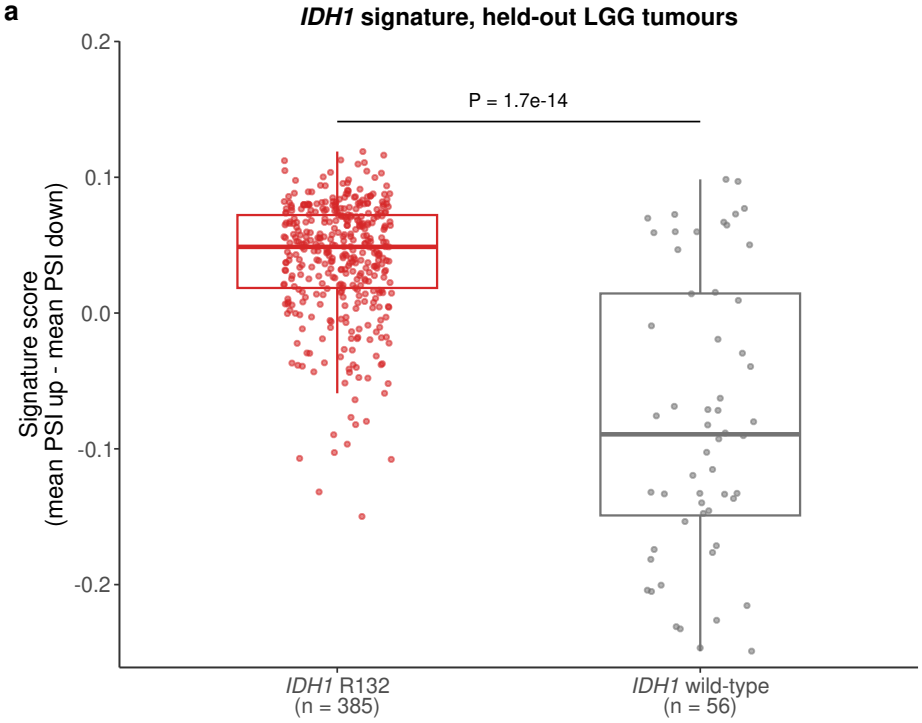
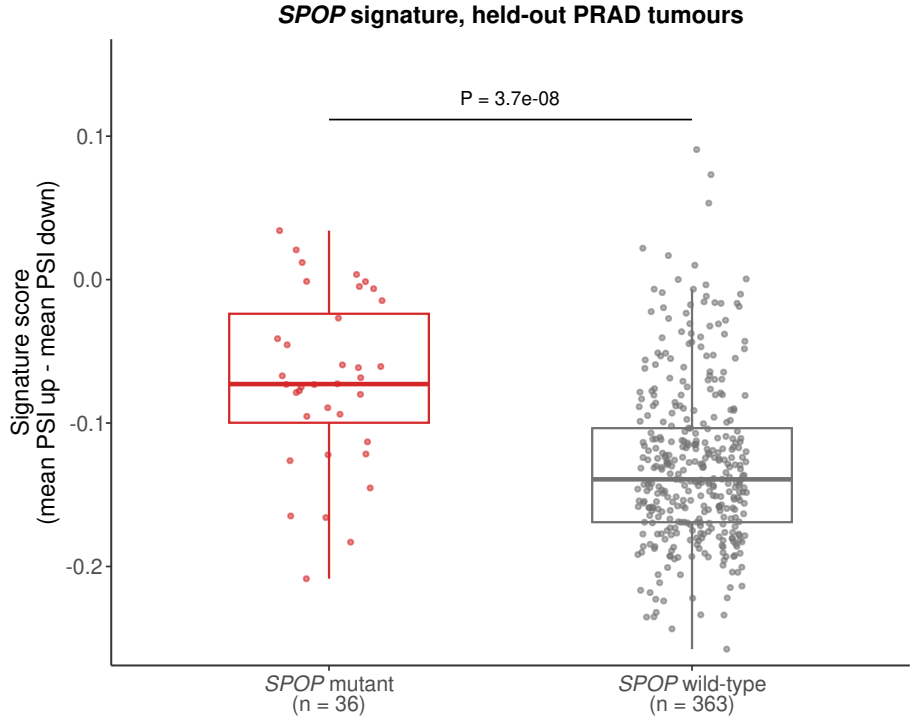
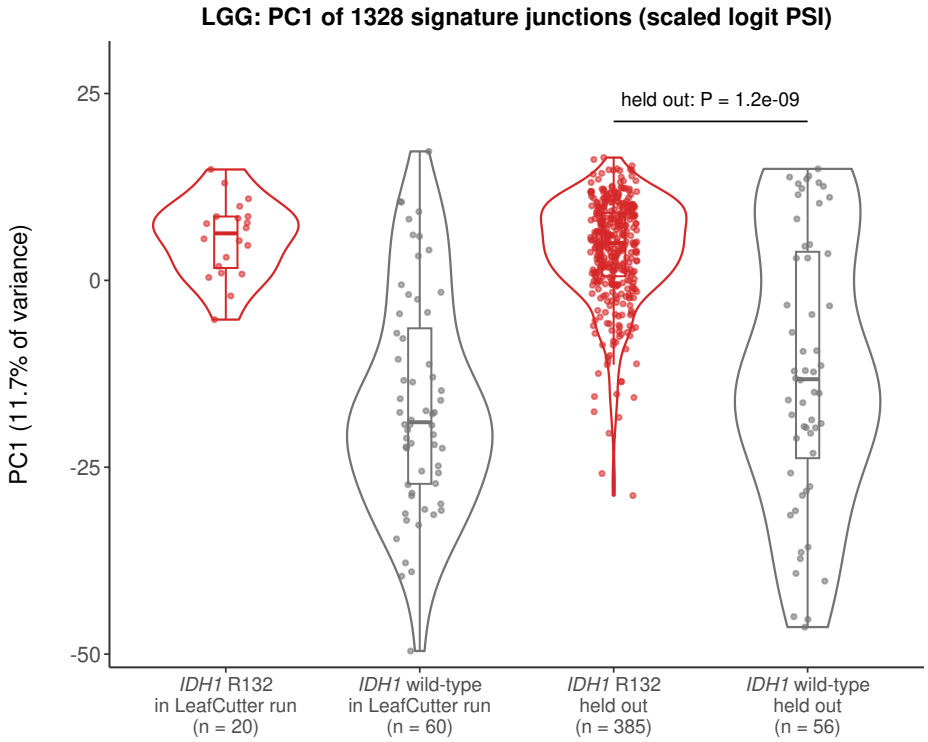
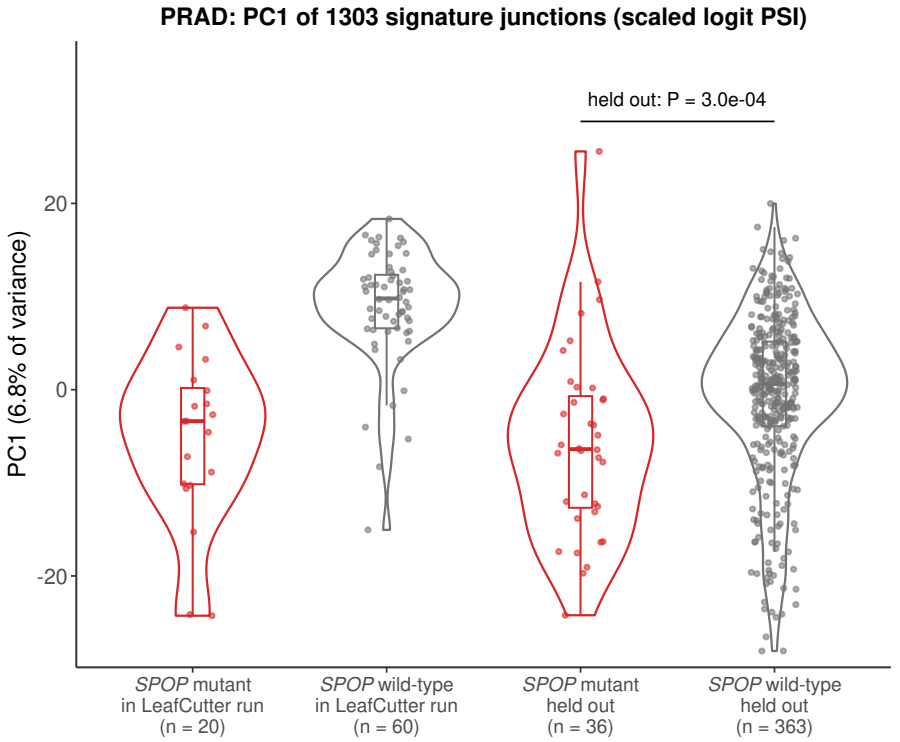
a

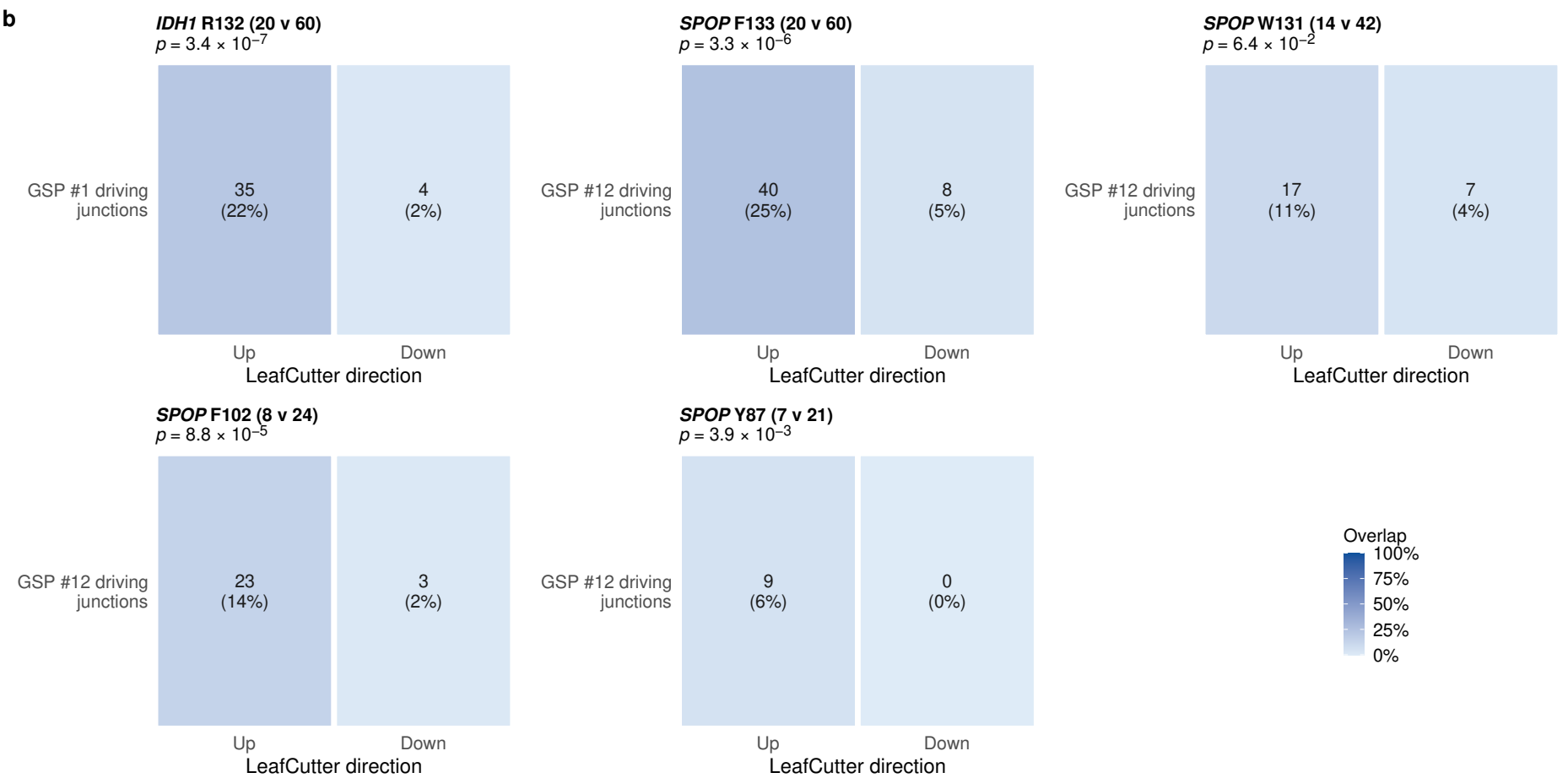

IDH1 signature, held-out LGG tumours
P = 1.7e-14
Signature score
(mean PSI up - mean PSI down)
0.2
0.1
0.0
-0.1
-0.2
IDH1 R132
(n = 385)
IDH1 wild-type
(n = 56)


SPOP signature, held-out PRAD tumours
P = 3.7e-08
Signature score
(mean PSI up - mean PSI down)
0.1
0.0
-0.1
-0.2
SPOP mutant
(n = 36)
SPOP wild-type
(n = 363)


LGG: PC1 of 1328 signature junctions (scaled logit PSI)
held out: P = 1.2e-09
PC1 (11.7% of variance)
25
0
-25
-50
IDH1 R132
in LeafCutter run
(n = 20)
IDH1 wild-type
in LeafCutter run
(n = 60)
IDH1 R132
held out
(n = 385)
IDH1 wild-type
held out
(n = 56)


PRAD: PC1 of 1303 signature junctions (scaled logit PSI)
held out: P = 3.0e-04
PC1 (6.8% of variance)
20
0
-20
SPOP mutant
in LeafCutter run
(n = 20)
SPOP wild-type
in LeafCutter run
(n = 60)
SPOP mutant
held out
(n = 36)
SPOP wild-type
held out
(n = 363)

b


***IDH1* R132 (20 v 60)**
$p = 3.4 \times 10^{-7}$
GSP #1 driving junctions
35 (22%)
4 (2%)
Up
Down
LeafCutter direction
***SPOP* F133 (20 v 60)**
$p = 3.3 \times 10^{-6}$
GSP #12 driving junctions
40 (25%)
8 (5%)
Up
Down
LeafCutter direction
***SPOP* W131 (14 v 42)**
$p = 6.4 \times 10^{-2}$
GSP #12 driving junctions
17 (11%)
7 (4%)
Up
Down
LeafCutter direction
***SPOP* F102 (8 v 24)**
$p = 8.8 \times 10^{-5}$
GSP #12 driving junctions
23 (14%)
3 (2%)
Up
Down
LeafCutter direction
***SPOP* Y87 (7 v 21)**
$p = 3.9 \times 10^{-3}$
GSP #12 driving junctions
9 (6%)
0 (0%)
Up
Down
LeafCutter direction
Overlap
100%
75%
50%
25%
0%